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Why the AI Boom Is About to Hit a “Flash” Wall

Jordi Visser

JVisser@22VResearch.com

22vresearch.com

Catalyst: Phison Electronics CEO KS Pua

The “Bubble” vs. The Buyer

The parabolic move in memory equities over the past year has triggered a familiar reflex across institutional portfolios. Historically, vertical price charts in NAND and DRAM have marked late-cycle peaks. In 2017–2018 and again in 2021, tight supply conditions drove sharp pricing spikes only to be followed by oversupply, margin compression, and severe equity drawdowns.

Given that history, it is rational for investors to ask whether the current move represents another cyclical blow-off top. But this cycle presents a materially different signal.

In prior peaks, suppliers were aggressively expanding capacity into rising prices. Inventory built across the channel. Pricing strength was often accompanied by promotional commentary from producers. When hyperscale or smartphone demand slowed, the system snapped back violently.

Today’s tone is different. The warning is not coming from a manufacturer attempting to push product into the channel. It is coming from companies struggling to secure allocation. The language has shifted from “pricing leverage” to “volume rationing.” In several instances, buyers report that even prepaid capital cannot guarantee supply. That distinction matters. Cyclical peaks are typically driven by demand elasticity; structural shortages are driven by supply inelasticity.

This is occurring at the same time that AI infrastructure is transitioning from concentrated training clusters to broader inference deployment. Training demand is episodic and project-based. Inference demand is distributed and persistent. That shift materially increases storage intensity per workload, not just in capacity, but in read/write frequency and durability requirements. Flash is no longer merely archival storage in this architecture; it becomes an active component of the compute stack.

Importantly, memory today is being consumed by buyers with vastly different price sensitivity than in prior cycles. In consumer electronics, NAND is a meaningful percentage of bill of materials. In AI servers, it is a small fraction of system cost. When allocation tightens, hyperscale buyers can absorb significant price increases without disrupting total system economics. That alters the traditional demand destruction mechanism that historically capped memory upcycles.

None of this eliminates cyclicity from semiconductors. Capacity will eventually expand. Pricing will eventually normalize. But the defining feature of 2026 is not speculative inventory build, it is infrastructure-level allocation stress in a market where supply elasticity is measured in years, not quarters. This is the core expression of a broader trade: long scarcity, the physics layer of silicon, power, and cooling that cannot scale on demand and short abundance, the software layer whose marginal cost assumptions depend on that physical infrastructure arriving on time.”

The 2,000 TWD Chair

In November 2025, Phison Electronics CEO KS Pua appeared on the prominent Taiwanese financial program *Era Money*. On that day, Phison’s stock was trading near 1,000 TWD. While Pua was already bullish then, predicting a decade-long

shortage, he returned to the same chair Friday February 13th with a message that had shifted from “optimistic” to “alarmist.”

With the stock price having doubled to 2,000 TWD in just three months, Pua was no longer just predicting a trend; he was describing a structural break in the global semiconductor value chain. **“In my 40 years in this industry,” he warned, “this only happens once in a lifetime.”**

The shift between these two interviews represents the move from theoretical demand to physical scarcity. In November, Pua argued that AI was a “necessity” and predicted a long-term storage squeeze. By February, the tone turned much darker. **He now cites a “physics wall” where supply cannot catch up regardless of capital.**

The Math of Scarcity

The transition from general-purpose compute to AI-native infrastructure radically changes the bill of materials. NVIDIA’s Vera Rubin GPU, expected to ship in volume by year-end, requires over 20 terabytes of high-speed SSD per unit. Even assuming mid-single-digit millions of units annually, that equates to roughly 200 exabytes of NAND demand. Global NAND production last year totaled approximately 1,100 exabytes. One architecture alone could absorb roughly 20% of global Flash output. Even if actual unit volumes fall short of this estimate, the directional imbalance remains material. Furthermore, this calculation excludes the secondary effect: the data these systems generate once deployed, training logs and inference traces, which create sustained write intensity and accelerate storage wear.

AI Is a Necessity and Only Three Years Old

Pua’s core argument is that AI is structural demand, not speculation. Historically, tech cycles lasted decades: PCs for 40-plus years, smartphones for over 20. AI, counting from late 2022, is barely three and a half years old. Meanwhile, the combined capital expenditure of the four major U.S. cloud providers is projected to exceed \$600 billion this year. This represents the early phase of what appears to be a multi-year infrastructure ramp. That spending is directed almost entirely toward data center buildouts: wafers, GPUs, networking, and storage. \$600 billion exceeds the GDP of most countries and we are just reaching the steepest part of the curve.

The Three-Year Prepayment Signal

The most telling data point is behavioral, not numerical. A major NAND producer (SanDisk/Kioxia) is now requiring three years of prepaid cash to secure supply. Even TSMC does not require such extreme terms from NVIDIA. This is not typical semiconductor cyclicality; it is seller’s-market discipline in a supply-constrained environment.

Phison itself must raise working capital materially to secure upstream allocation. It is critical to note that Phison acts as a primary buyer of raw NAND flash, which they then process into high-value controllers and modules. When a buyer who processes billions of dollars of silicon says the market is sold out, it carries more weight than a seller trying to move inventory. Suppliers are not short on cash, they are short on product. One supplier’s response was blunt: “We don’t need money. We don’t have any product.”

Demand Drivers: Sequencing the Waves

Demand Wave	Status (Feb 2026)	Confidence Level	Timeline
U.S. Cloud / CSP	Active — \$600B+ capex	High (Contracted)	Accelerating through 2027+
China Cloud	Pre-ramp	Emerging (Sovereign AI)	Ramp begins 2026–2027
Edge AI / On-Device	Nascent	Contingent (Device Refresh)	Consumer rollout H2 2026
Institutional / Education	Early adoption	High (Budgeted)	2026–2028 deployment

The Season of Sacrifice

In AI servers, storage represents roughly 5–6% of the bill of materials; price increases are absorbable. In consumer electronics, storage can exceed 20% of the cost. In a \$300 television, storage cost increases from \$1.50 to \$20 can destroy product viability. This dynamic effectively reallocates scarce silicon from discretionary consumer demand to mission-critical AI infrastructure, altering global electronics trade flows. When hyperscalers can pay triple for NAND and consumer OEMs cannot pass through cost, allocation is inevitable. The result is compressed unit shipments for smartphones and a shift toward repair over replacement in the mass market.

Processors vs. Growers: Who Wins?

Pua describes Phison as a “rice processor,” not a grower. The “growers”, Samsung, SK Hynix, Micron, produce raw NAND and suffer commodity volatility. “Processors” add value through controller intelligence, firmware, and system design. In scarcity, efficiency arbitrage dominates. Phison has maintained approximately 30% gross margins through two and a half boom-bust cycles by extracting higher performance and durability from constrained NAND supply, same raw material, different value. In a shortage, the growers capture commodity price. The processors capture margin on intelligence. That distinction is where the alpha resides, and it is the logic underpinning the Storage Platforms tier of the investment matrix below.

Strategic Investment Matrix: 2026–2027

Vertical	Representative Exposure	Structural Edge
NAND Equipment	LRCX, AMAT	Gatekeepers of 400+ layer manufacturing tools.
Connectivity	MRVL, ALAB, AVGO	Bandwidth must scale with storage density.
Storage Platforms	PSTG, 8299.TW	Efficiency arbitrage in a world of scarce silicon.
Power & Cooling	VRT, VICR	AI servers require liquid cooling and 800V DC.

Conclusion: Survival as Strategy

The “Flash Wall” is not a temporary peak; it is a multi-year repricing of infrastructure. Memory has historically traded as a cyclical commodity. In an AI-constrained world, it begins to trade as strategic infrastructure, a multiple shift, not just an earnings cycle. For those looking at the parabolic charts and preparing to sell, the physical reality in the factories suggests a different conclusion: we are in the early innings of a fundamental resource reallocation. The structural imbalance between exponential AI storage intensity and physically constrained NAND expansion will likely persist through 2027, with risk skewed to upside pricing pressure. Whether prices overshoot in the short term is a tactical question. The supply inelasticity is the strategic one and it reinforces the foundational thesis: long scarcity, short abundance. Every software valuation built on the assumption of infinite, cheap inference capacity is implicitly short the physical constraints documented in this report. The companies that own the scarce layers, equipment, storage architecture, power delivery, sit on the other side of that trade.

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